ZHOU Yang

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Undergraduate, Class of 2023
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Education

B.S. in Computational Mathematics **Sep 2023 – Jul 2027 (expected)**
Wuhan University, Wuhan, China
Major GPA: 3.96/4.00
Relevant Coursework: Mathematical Analysis, Advanced Algebra, Analytic Geometry, Probability, Mathematical Statistics, Numerical Linear Algebra, Numerical Analysis, Optimization Methods, Real Analysis, Partial Differential Equations, Numerical Methods for Differential Equations, Functional Analysis.
Honors:

- National Scholarship (China), 2025
- Merit Student Honor (Highest Distinction), Wuhan University, 2025
- First-Class Merit Scholarship, Wuhan University, 2024 & 2025
- Merit Student Honor, Wuhan University, 2024 & 2025

Competitions & Awards

- **Third Prize**, 16th *“Ziqiang Cup” College Students’ Extracurricular Scientific and Technological Innovation Contest*, Wuhan University, Mar 2025
- **Honorable Mention**, *The Mathematical Contest in Modeling (MCM)® / Interdisciplinary Contest in Modeling (ICM)® (COMAP)*, Jan 2025
- **First Prize**, *Asia and Pacific Mathematical Contest in Modeling (APMCM)*, Nov 2024
- **First Prize (Hubei Province), Mathematics Major Track**, *The Chinese Mathematics Competitions for College Students (CMC)*, Nov 2024
- **Second Prize (Hubei Province)**, *“Higher Education Press Cup” Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)*, Sep 2024
- **First Prize**, *“Huazhong Cup” College Students Mathematical Modeling Challenge*, Apr 2024

Projects

Algorithms for Computing Multiple Real Roots of Polynomials and Smooth Functions
Oct 2025 – Dec 2025
Course Project in Numerical Analysis; in collaboration with Ren Yankun (Wuhan University)

- Conducted a literature review and focused on two classes of polynomial root-finding methods: the companion-matrix eigenvalue method and the Sturm-sequence method.
- Motivated the necessity of working with orthogonal polynomial bases and developed a root-finding approach based on Chebyshev polynomials.
- Discussed difficulties in root finding for smooth functions and studied the Boyd–Battles method based on polynomial interpolation and companion-matrix eigenvalues.
- Ran numerical experiments on polynomials and smooth functions with multiple real roots; verified correctness and analyzed stability.

Linear System Solvers for Finite-Difference Discretizations of Two-Point Boundary Value Problems
May 2025
Course Project in Numerical Linear Algebra

- Studied solver behavior on a model problem: a 1D convection-diffusion boundary value problem with a small parameter, discretized by finite differences.
- Compared direct methods (Gaussian elimination, Cholesky, Thomas algorithm) and iterative methods (Jacobi, Gauss–Seidel, SOR, Conjugate Gradient), analyzing runtime, accuracy, and (for iterative methods) convergence behavior.
- Discussed modeling error, limiting behavior of the two-point BVP, and the impact of finer mesh resolutions.

Additional Projects

- Simulation and pattern formation of the Gray–Scott reaction-diffusion system, Course Project in Computational Physics, Jan 2026
- Numerical experiments for unconstrained and constrained optimization, Course Project in Optimization Methods, Dec 2025 – Jan 2026
- Talk: Algorithms for nonlinear least squares (Presenter), Continuous Optimization Reading Group, Nov 2025
- Beitai Tianyuan Scientific Computing Training Camp group project (scientific plotting), Team Project, Jul 2025

Research Experience

Implicit Bias of Neural Networks for Solving Differential Equations Jul 2025 – Present

Advised by Prof. Zhao Xiaofei (Wuhan University) and Prof. Luo Tao (Shanghai Jiao Tong University)

- Systematically studied fundamentals of deep learning, neural-network approaches for differential equations (e.g., PINNs, Deep Ritz, DLAM), and representative literature on implicit bias (e.g., frequency principle and condensation phenomena).
- Experimentally investigated how different loss formulations affect hyperparameter sensitivity in differential-equation tasks; compared PINN and DLAM from the perspectives of loss landscapes and training dynamics, and summarized key empirical findings and trends.
- Combined theory and experiments to study condensation phenomena, including initialization conditions, impacts on solution quality, and potential mechanisms.

Theory and Computation of Rational Approximation Sep 2024 – Jun 2025

Provincial Undergraduate Innovation Training Program

In collaboration with Huang Nanqiao (Wuhan University); advised by Prof. Lü Xiliang (Wuhan University)

- Motivated the need for rational approximation: in applications (e.g., signal processing), functions with poles are common; polynomial or trigonometric approximations may fail, while rational functions can provide accurate approximations.
- Studied classical methods (Padé approximation, least-squares rational approximation) and modern algorithms (e.g., AAA); discussed underlying ideas, convergence, and stability, and compared practical performance via numerical experiments.
- Explored combining rational approximation with neural networks by implementing neural networks with rational activation functions; validated feasibility on simple tasks (e.g., function fitting) and discussed potential for more complex problems.

Skills

Python, C, MATLAB, R

Languages: English (CET-4: 595; CET-6: 503); Mandarin Chinese (native)